

Personalized AI Tutor - An AI-Powered Learning Recommendation System

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Introduction

Students today have access to endless learning resources, but finding the right one is difficult. Most platforms give the same recommendations to everyone, ignoring individual needs.

A personalized AI Tutor solves this problem. It asks students about their skill level, interests, available time, past performance, and goals. Using Machine Learning, it recommends:

- **What topic** to study next
- **How difficult** the topic should be
- **What format** to use (video, article, interactive, or book)

Problem Statement

Students face three main problems:

Problem	Description
No Personalization	Everyone gets the same recommendations
No Guidance	Students don't know what to study next

No Explanation	Students don't understand why a recommendation is made
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Our system solves all three problems by providing personalized, explainable recommendations.

Dataset

We used a synthetic dataset with **2000 student records**.

Input Features

Feature	Values
Skill Level	Beginner, Intermediate, Advanced
Interest	Python, Web Development, Data Science, ML, Mobile
Time Available	1-2 hrs, 2-4 hrs, 4-6 hrs, 6+ hrs
Past Performance	Score from 50 to 100
Learning Goal	Career, Hobby, Academic, Certification

Output Predictions

Output	Values
Recommended Topic	Topic name based on interest and level
Difficulty	Easy, Medium, Hard
Resource Type	Video, Article, Interactive, Book

The dataset was created using educational rules. For example:

- Beginners always get basic topics
- High performers get advanced topics
- Time availability determines resource type

Methodology

Algorithm Used

We used **Random Forest Classifier** for three reasons:

1. **Good Accuracy** — Performs well on our data
2. **Feature Importance** — Shows which factors matter most
3. **Handles Mixed Data** — Works with both text and numbers

How It Works

Student Input → Encode Text → Random Forest Models → Recommendations

We trained **three separate models**:

1. One for predicting Topic
2. One for predicting Difficulty
3. One for predicting Resource Type

Training Process

- 80% of data used for training
- 20% used for testing
- 100 decision trees in the Random Forest

Implementation

Technology Used

Technology	Purpose
Python	Programming language
Streamlit	Web interface
Scikit-learn	Machine Learning
Plotly	Charts
Pandas	Data handling

System Features

- **Input Form:** Students enter their profile details
- **Recommendations:** Three outputs with confidence scores
- **Explanation:** Natural language explanation
- **Chart:** Shows which factors influenced the decision

Results

Model Accuracy

Prediction	Accuracy
Topic	88.3%
Difficulty	93.3%
Resource	100%

Overall	93.9%
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Feature Importance

Factor	Importance
Past Performance	32.2%
Interest	30.6%
Skill Level	24.3%
Time Available	6.9%
Goal	6.0%

Key Finding: Past Performance and Interest have the strongest influence on recommendations.

Model Comparison

We compared Random Forest with two other models:

Model	Overall Accuracy
Random Forest	93.9%
Decision Tree	89.9%
Logistic Regression	85.2%

Conclusion: Random Forest performs best, so we selected it.

Explainability

Natural Language Explanation

Example:

"Hi Samad! You're a Beginner in Web Development. With 67/100 performance, we recommend HTML/CSS Basics with Easy difficulty. With 1-2 hours available, Article-based learning is best."

Feature Importance Chart

A bar chart shows which factors influenced the recommendation. Past Performance and Interest have the highest bars.

Limitations

Limitation	Explanation
Synthetic Data	Data is not from real students
Limited Features	Only 5 input features
No Tracking	System doesn't track progress
Limited Topics	Only 5 interest areas

Future Work

1. Use real student data
2. Add more features (age, past experience)
3. Track student progress

4. Build a mobile app

Conclusion

We built an AI-powered recommendation system that:

- Provides personalized learning recommendations
- Achieves 93.9% accuracy
- Explains recommendations in plain language
- Shows which factors influence decisions

The system demonstrates how Machine Learning can help students find the right learning path.